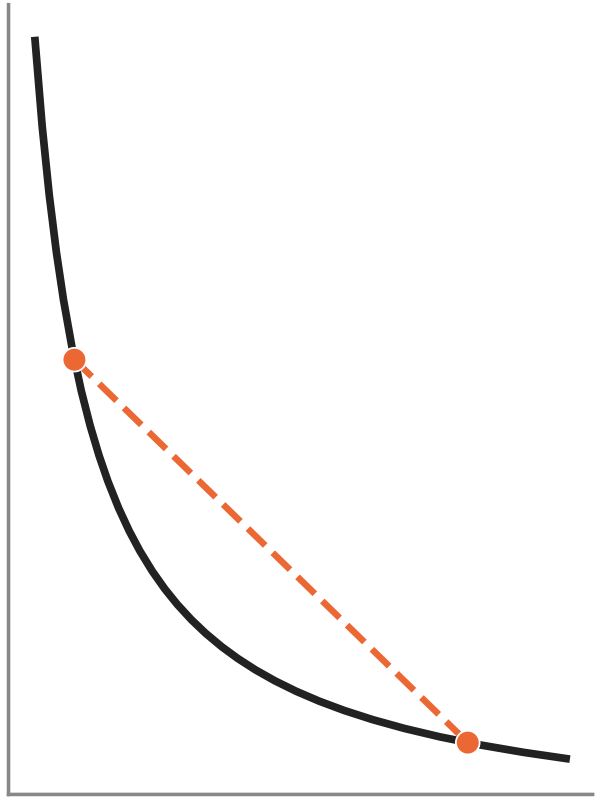
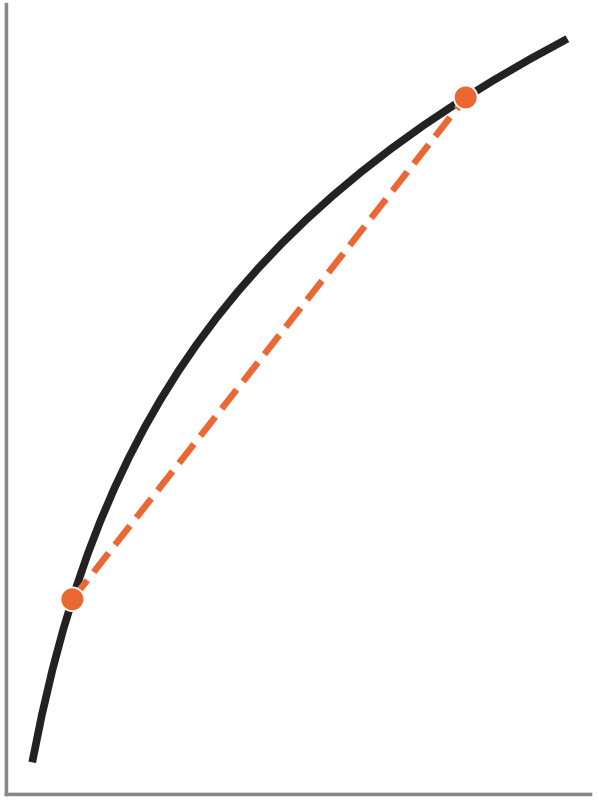
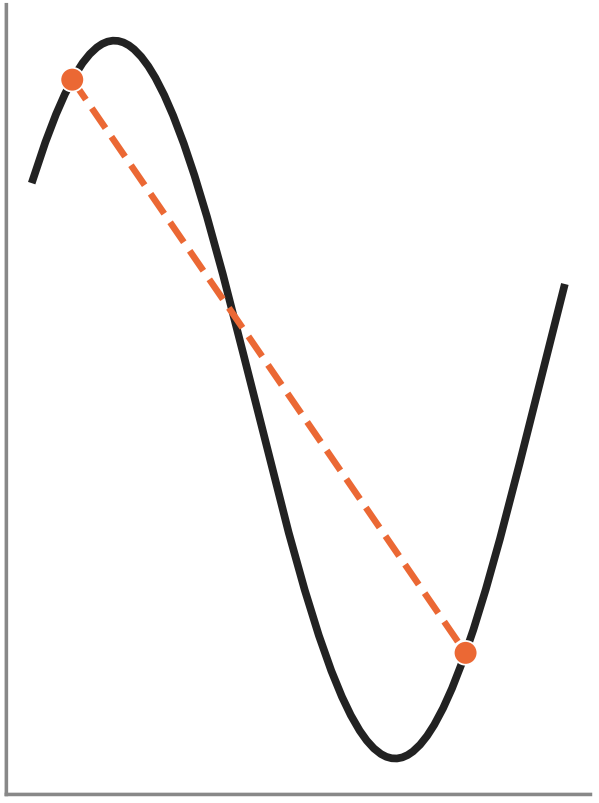
Convex: $1/x$



Concave: $\log x$



Neither: $\sin 2.2x$